MILITARY SPECIFICATION

BATTERY, DRY, BA-381/U

This specification has been approved by the
Bureau of Naval Weapons, Department of the Navy.

1. SCOPE

1.1 Scope. - This specification covers the detail requirements for the Battery, Dry, BA-381/U, a Category I battery.

2. APPLICABLE DOCUMENTS

2.1 The following specification and standard of the issue in effect on date of invitation for bids, form a part of this specification, to the extent specified herein.

SPECIFICATION

MILITARY

MIL-B-18 Batteries, Dry.

STANDARD

MILITARY

MIL-STD-8 Dimensioning and tolerancing

(Copies of specifications and standards required by contractors in connection with specific procurement functions should be obtained from the procuring activity or as directed by the contracting officer.)

3. REQUIREMENTS

3.1 Requirements shall be in accordance with Military Specification MIL-B-18, and as specified herein.

3.1.1 General.

Nominal voltage - - - - - - - - - 1.5 volts
Usual number and type of cells - 1 "F" cell
Terminals - - - - - - - - - - - - Socket (See Figure 1)
Maximum weight - - - - - - - - 8 ounces
Dimensions - - - - - - - - - - - See figure 1

3.1.2 Electrical.

Open circuit voltage - - - - - - 1.67 volts max.
Closed circuit voltage through 5 ohms for 5 seconds - 1.41 volts min.

3.1.3 Magnetic effect. - This battery shall meet all magnetic effect requirements of MIL-B-18. Resistance for the magnetic effect under load shall be 16.7 ohms.

3.1.4 Environmental.

Vibration test I
Mechanical shock test I

3.1.5 Capacity. - The battery shall be discharged for 5 hours per day, 5 days per week. This cycle shall be repeated continuously. Minimum capacity values shall be as indicated below:

Test	Discharge load (ohms)	Test-End Voltage (volts)	Min. capacity value (hours)
D	20	0.9	130
T	20	0.9	100
DD	20	0.9	130

3.1.6 Electrolyte leakage. - Following IEL and DEL storage, the battery shall be discharged through 5 ohms for 24 hours. DEL storage shall be 12 months.

3.1.7 Labeling. - Labeling shall be in accordance with Military Specification MIL-B-18 and Figure 1.

4. QUALITY ASSURANCE PROVISIONS

4.1 Preproduction tests. - Five additional sample batteries shall be submitted to the activity designated in the contract. (See 6.1)

4.2 Test equipment. - Test Set TS-183( )/U shall not be used in testing this battery.

FSC 6135

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4.3 Examinations and tests.

(a) Open-circuit voltage (See 4.3.1)
(b) Closed-circuit voltage (See 4.3.1 and 4.3.3)
(c) Visual and mechanical examinations
(d) Vibration test I
(e) Mechanical shock test I
(f) Magnetic effects tests
(g) Storage (See 4.3.1)
(h) Capacity tests D, T, and DD
(i) Electrolyte leakage

4.3.1 Prior to sampling for Group A inspection, each battery manufactured shall be tested to determine that the requirements for Open-circuit voltage and Closed-circuit voltage have been met.

4.3.1.1 Closed-circuit voltage shall be read at end of five second test interval.

4.3.2 Storage for capacity tests. - Batteries as received (in unit packages) shall be stored as follows:

For D Test	12 months at 70°F, 50% RH
For T Test	3 months at 113°F, 50% RH, and 48 hours at 70°F, 50% RH
For DD Test	24 months at 70°F, 50% RH

4.3.3 After storage and just prior to D and T capacity discharge, the batteries shall be subjected to the closed-circuit voltage test, at 70°F.

4.4 Inspection of preparation for delivery shall be made to ascertain conformance to Section 5.

5. PREPARATION FOR DELIVERY

5.1 Packaging and packing shall be in accordance with Group I requirements of MIL-B-18, and as follows:

Unit package	15 batteries
Approx. inside dimensions	4-1/8 x 6-3/4 x 4-1/8 inches
Arrangement	3-5-1
Shipping container	6 unit packages
Approx. inside dimensions	9 x 7-1/2 x 13 inches
Arrangement	2-1-3

5.2 Special marking. - The following special marking shall be prominently displayed on each shipping container

WARNING

FRAGILE, HANDLE WITH CARE. Maintain temperature below 45°F while handling for shipment, during transit or during storage. INJURIOUS TO CONTENTS if temperature rises above 45°F.

6. NOTES

6.1 Ordering data. - Procurement documents should also specify:

(a) Activity to which additional sample batteries shall be submitted.

(b) The following statements:

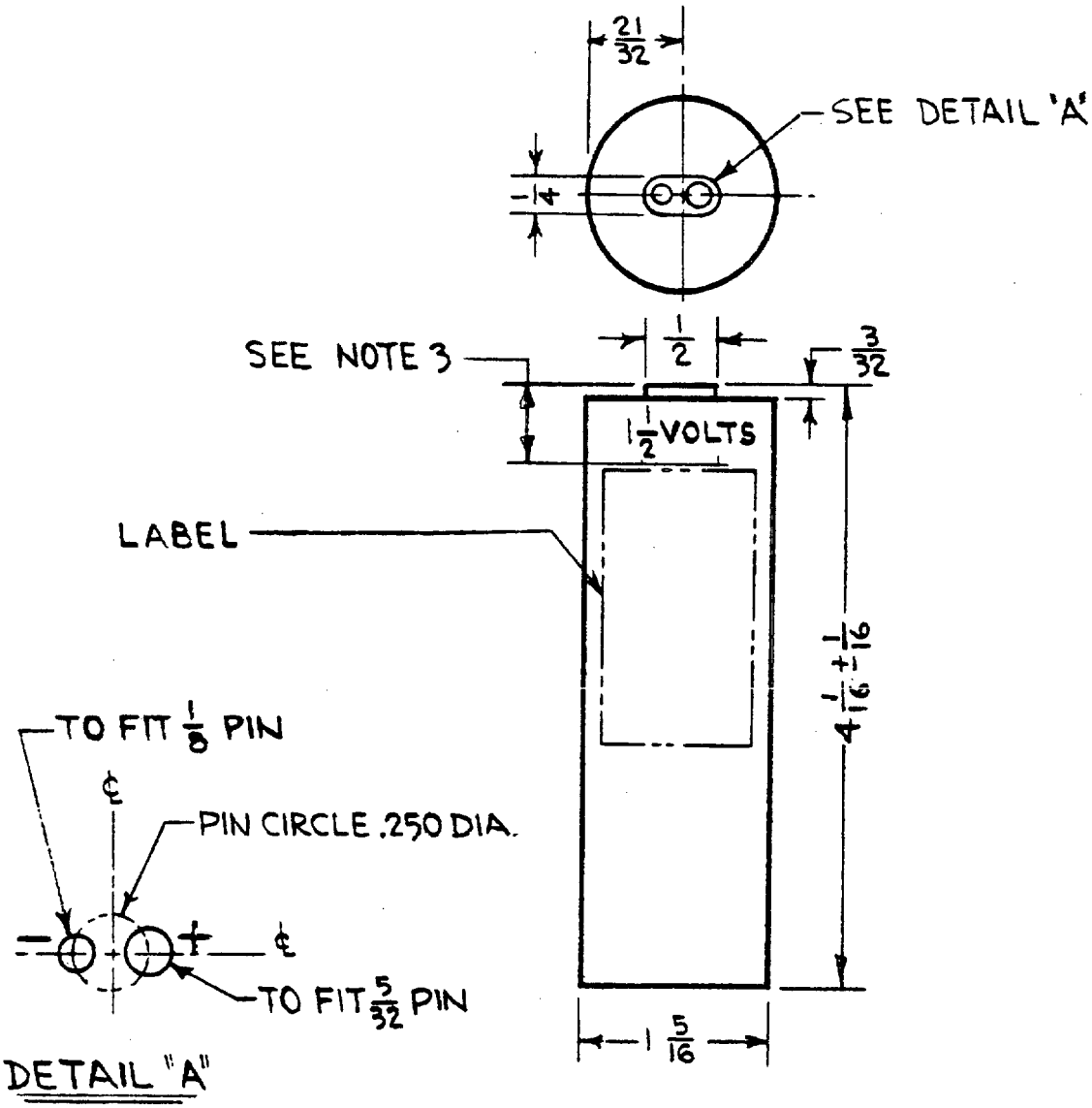
(1) When shipping batteries, carrier shall maintain temperature below 45°F.

(2) Batteries shall not be kept in an unrefrigerated condition for more than seven days prior to shipment.

Notice. - When government drawings, specifications, or other data are used for any purpose other than in connection with a definitely related Government procurement operation, the United States Government thereby incurs no responsibility nor any obligation whatsoever; and the fact that the Government may have formulated, furnished, or in any way supplied the said drawings, specifications or other data is not to be regarded by implication or otherwise as in any manner licensing the holder or any other person or corporation, or conveying any rights or permission to manufacture, use, or sell any patented invention that may in any way be related thereto.

Preparing activity:
Navy-Bureau of Naval Weapons

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NOTES:

1. ALL DIMENSIONS ARE IN INCHES.
2. UNLESS OTHERWISE SPECIFIED, TOLERANCES ARE ±1/32 ON FRACTIONS AND ±.005 ON DECIMALS.
3. SOCKET WELL DEPTH 1/2" MINIMUM.
4. BATTERY SHALL BE SUPPLIED WITHOUT WATERPROOFING.

FIGURE 1

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SPECIFICATION ANALYSIS SHEET
NAVWEPS FORM 4121/3 (8-62)

FORM APPROVED
BUDGET BUREAU NO. 45-R309

INSTRUCTIONS

This sheet is to be filled out by Government or contractor personnel involved in the use of this specification in procurement of products for ultimate use by the activity shown on the reverse of this sheet.

This sheet is provided for obtaining information which will aid the activity shown in insuring that through use of this specification suitable products can be procured with a minimum amount of delay and at the least cost.

Comments and the return of this sheet will be appreciated.

Fold on dotted lines on reverse side, staple in corner, and send to the activity shown.

SPECIFICATION *(No. and abbrev. title)*

ORGANIZATION	CITY	STATE

CONTRACT NUMBER	QUANTITY OF ITEMS PROCURED	DOLLAR AMOUNT
		$

MATERIAL PROCURED UNDER A DIRECT GOVERNMENT CONTRACT ☐ OR A SUBCONTRACT ☐

1. HAS ANY PART OF THIS SPECIFICATION CREATED PROBLEMS OR REQUIRED INTERPRETATION IN PROCUREMENT USE?
(If so, give paragraph number and wording, and recommendations for correcting the deficiencies.)

2. COMMENTS ON ANY SPECIFICATION REQUIREMENT CONSIDERED TOO RIGID

3. IS THE SPECIFICATION RESTRICTIVE?
☐ YES ☐ NO IF YES, IN WHAT WAY?

4. REMARKS *(Attach any pertinent data which may be of use in improving this specification. If additional papers are attached, send this form and papers in an envelope. This form is addressed for use in window envelope when appropriate.)*

SUBMITTED BY *(Print name and title)*	DATE

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